TEST YOUR KNOWLEDGE

1 True or false. Correct the false statements:

a) The S.I unit of angular momentum is $kg.m^2.s$

b) The angular momentum with respect to a fixed axis is always positive.

c) The moment of an external force applied on a solid rotating about a fixed axis changes its angular momentum.

d) A solid rotating about a fixed axis is subjected to external forces whose resultant moment with respect to this axis is zero. The angular momentum of the solid is then zero.

e) Two particles, rotating about a fixed axis in the same direction and with the same angular velocity, have the same angular momentum with respect to this axis. The angular momentum of the system formed by these particles is zero.

f) The derivative, with respect to time, of the angular momentum of a solid with respect to a fixed axis is equal to the sum of the external forces applied on the solid.

2 Choose the best answer:

A) The moment of inertia of a solid with respect to an axis of rotation (Δ) is 1 $kg.m^2$. The angular momentum of the solid rotating about (Δ) with the speed of 10 r. p. s in the positive direction of rotation is

a. 10 $kg.m^2/s$.

b. 62.8 $kg.m^2/s$.

c. -62.8 $kg.m^2/s$.

B) If the sum of moments of external forces exerted on a solid rotating about a fixed axis (Δ) is constant at any time, the angular momentum of the solid is then

a. constant.

b. variable.

c. zero.

C) A horizontal metallic rod rotates about a vertical axis (Δ) passing through its midpoint with a constant angular velocity. We heat the rod.

a. Its moment of inertia with respect to (Δ) decreases.

b. Its angular momentum with respect to (Δ) varies.

c. Its angular velocity decreases.

D) A man is standing on a large horizontal disk which can rotate without friction about its axis (Δ). The system (disk, man) is initially at rest. The man moves in a clockwise direction along the periphery of the disk. The disk

a. remains at rest.

b. rotates in a clockwise direction.

c. rotates in a counter-clockwise direction.

APPLY YOUR KNOWLEDGE

3 A hoop, of mass m and radius r, rotates about its axis with a speed of N r.ps.

a) Write the expression for the angular momentum of the hoop with respect to (Δ) in terms of m, r, and N.

b) What is the angular momentum, with respect to (Δ), of the center of gravity of the hoop, assuming that it carries the mass m?

c) Is it true that the angular momentum of

the system, with respect to a fixed axis, is equal to that of its center of gravity with respect to the same axis?

4 A particle rotating about a fixed axis (Δ) is subjected to only one force whose line of action meets (Δ). Is the angular momentum of the particle conserved? Justify your answer.

5 A hoop, of mass 200 g and radius 20 cm, rotates about its axis (Δ) with a speed of 10 r. p. s in the negative direction. Determine its angular momentum with respect to (Δ).
(Bear in mind that the moment of inertia of a hoop of mass m and radius r with respect to its axis is $I = mr^2$).

6 A system S is formed of two particles A_1 and A_2 rotating in opposite directions about the same axis (Δ). The moments of inertia of A_1 and A_2 with respect to (Δ) are respectively $I_1 = 0.1$ kg.m² and $I_2 = 0.3$ kg.m², while their respective angular velocities are and $\theta'_1 = 2$ rd/s and $\theta'_2 = -4$ rd/s. Determine the angular momentum of S with respect to (Δ).

7 A boy standing at the center of a horizontal turntable has his arms stretched and holds identical loads with each hand (Figure 3.10). The turntable rotates without friction about its axis with 0.50 r.p.s.
What will its speed be when the boy puts his hands down (Figure 3.11)? We suppose that the moment of inertia with respect to the axis of the system boy-turntable is 13.10 kg.m² (for outstretched arms) and 5.225 kg.m² (for hands down).

8 A horizontal disk of mass m = 500 g and radius r = 40 cm rotates without friction about its axis with a constant speed of 2 r.p.s. We suddenly put a particle of mass 100 g on the disk at 20 cm from its axis (Figure 3.12). Calculate the new speed of the disk.
Given $I_{disk} = \frac{mr^2}{2}$.

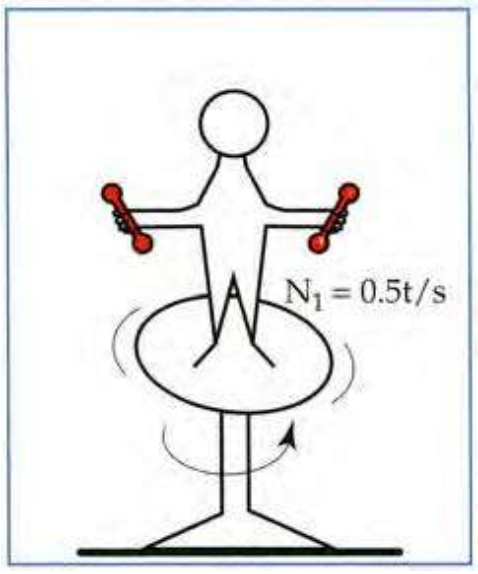

Figure 3.10

Figure 3.11

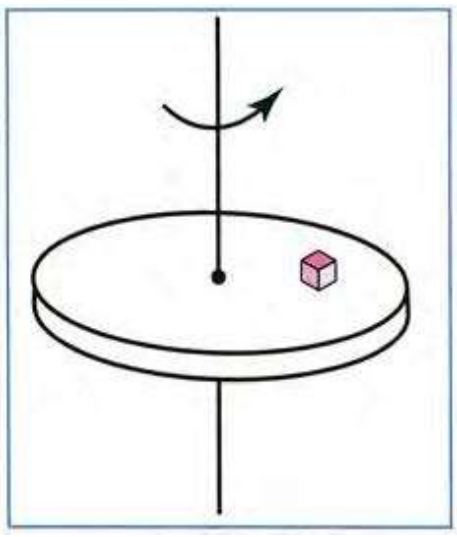
Figure 3.12

9 Two flywheels R_1 and R_2 rotate in two opposite directions about the same axis of rotation with a speed of 2 r.p.s. Their moments of inertia with respect to this axis are respectively $I_1 = 2.27$ kg.m² and $I_2 = 3.52$ kg.m². R_1 and R_2 are clutched together. Determine their common speed.

10 A block B of mass m = 100 g can slide without friction on a horizontal rod CD of mass m′ = 200 g and of length L = 50 cm. CD is coupled to an axis (Δ) passing through its midpoint O (Figure 3.13). B is attached at I to the axis by means of an inextensible stretched string of negligible mass and of length IO = 15 cm. The whole system rotates without friction about (Δ) with a speed of 2 r.p.s. The string is cut and B sticks to

extremity D. Calculate the new speed of the system. Given $I_{rod} = \frac{m'L^2}{12}$.

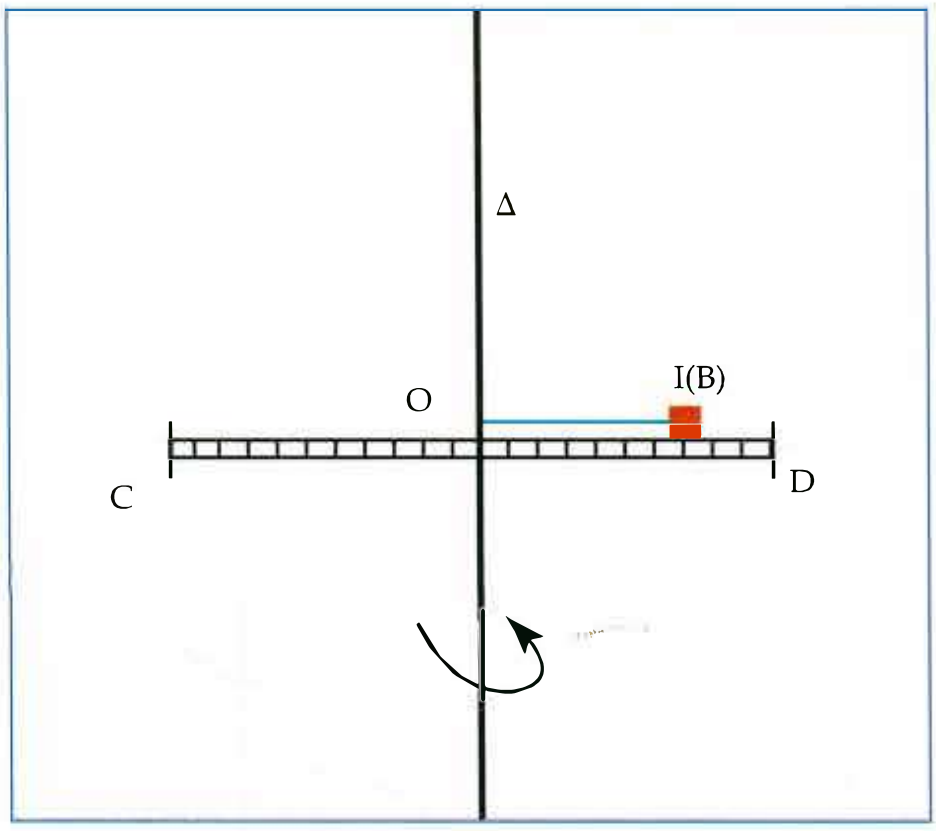

Figure 3.13

11 A flywheel, of radius r = 25 cm and moment of inertia I = 3.2 kg.m² with respect to its axis (Δ), rotates by means of a driving belt driven by an electric motor (Figure 3.14). Starting from rest, the flywheel is uniformly accelerated and reaches the speed of 2 r. p. s in 20 s.

a) Find the expressions for the angular velocity and angular momentum [with respect to (Δ)] of the flywheel during this starting phase.

b) Apply the theorem of angular momentum to calculate the tension F_1 of the belt. (F_2 is ignored with respect to F_1)

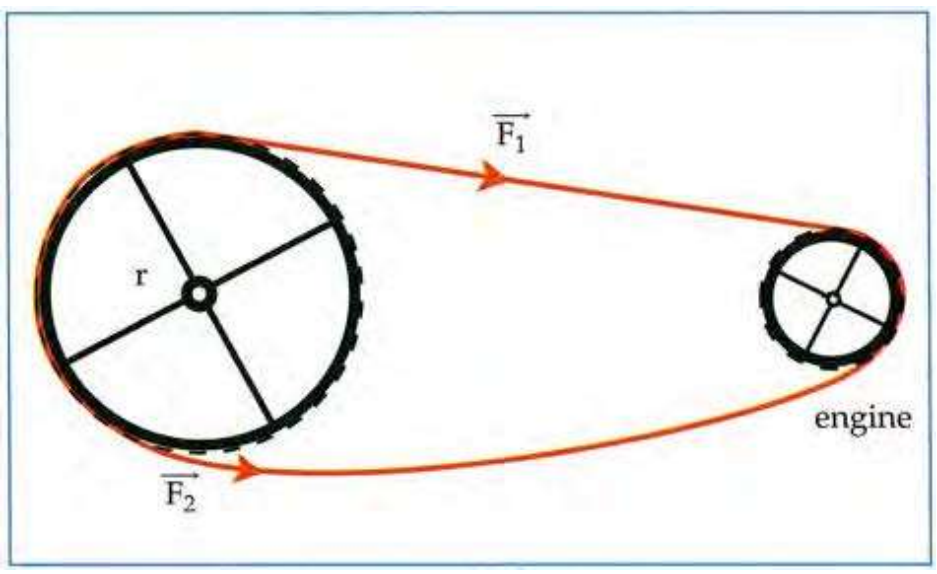

Figure 3.14

12 A hoop of radius r = 20 cm and mass m =150 kg can rotate about its axis. The hoop starts from rest at t = 0 s under the action of a tangential force $\vec{F}$ of constant magnitude F = 45 N. Calculate

a) its angular acceleration. Given $I_{hoop} = mr^2$.

b) its angular velocity at instant t_1= 4 s and the number of revolutions performed during these four seconds.

c) the work done by $\vec{F}$ during these four seconds. Verify that this work is equal to the kinetic energy of the hoop at t_1 = 4 s.

13 A thin piece of string of negligible mass is wound around the periphery of a disk D of radius R = 5 cm and of mass m = 3.2 kg. The disk is initially at rest. The string is pulled for 2 s with a constant force $\vec{F}$ tangent to the circumference of the disk (Figure 3.15). The disk rotates then without friction about its axis with a speed of 8 r.p.s. $I_d = \frac{mR^2}{2}$.

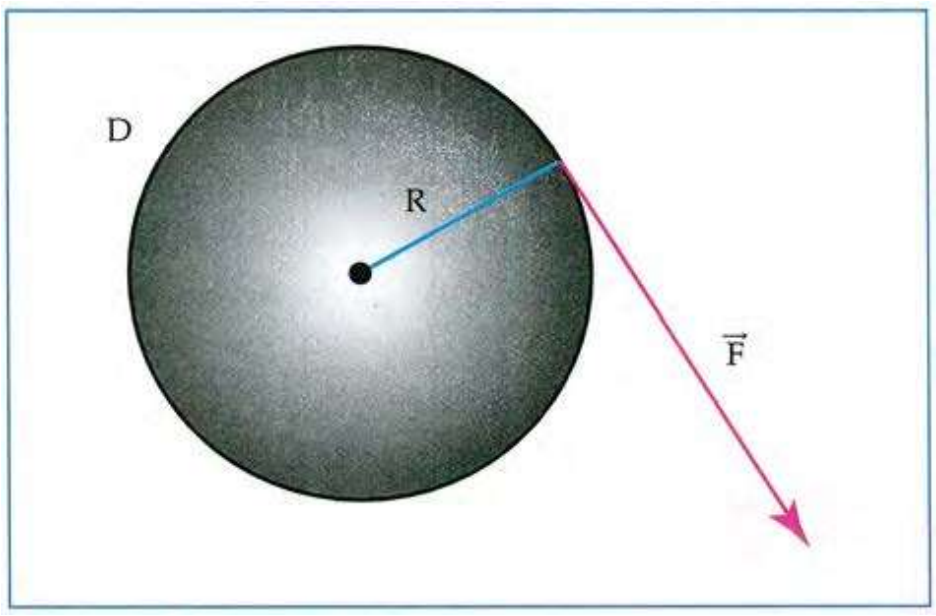

Figure 3.15

a) Apply the theorem of angular momentum to study the nature of the motion of the disk during the start.

b) Calculate the magnitude of $\vec{F}$.

c) A braking couple of moment M = − 0.2 m.N is applied to the disk for $t \geq 2$ s. Determine, by applying the theorem of angular momentum, the time needed for the disk to stop. Calculate the number of revolutions performed during the stopping time.

14 A particle A of mass m rests on a smooth horizontal table and is connected to a thin string which passes through a hole made in the table (Figure 3.16). The particle is given a uniform circular motion of radius R and angular velocity θ'_1 on the table, while an observer holds the other end of the string.

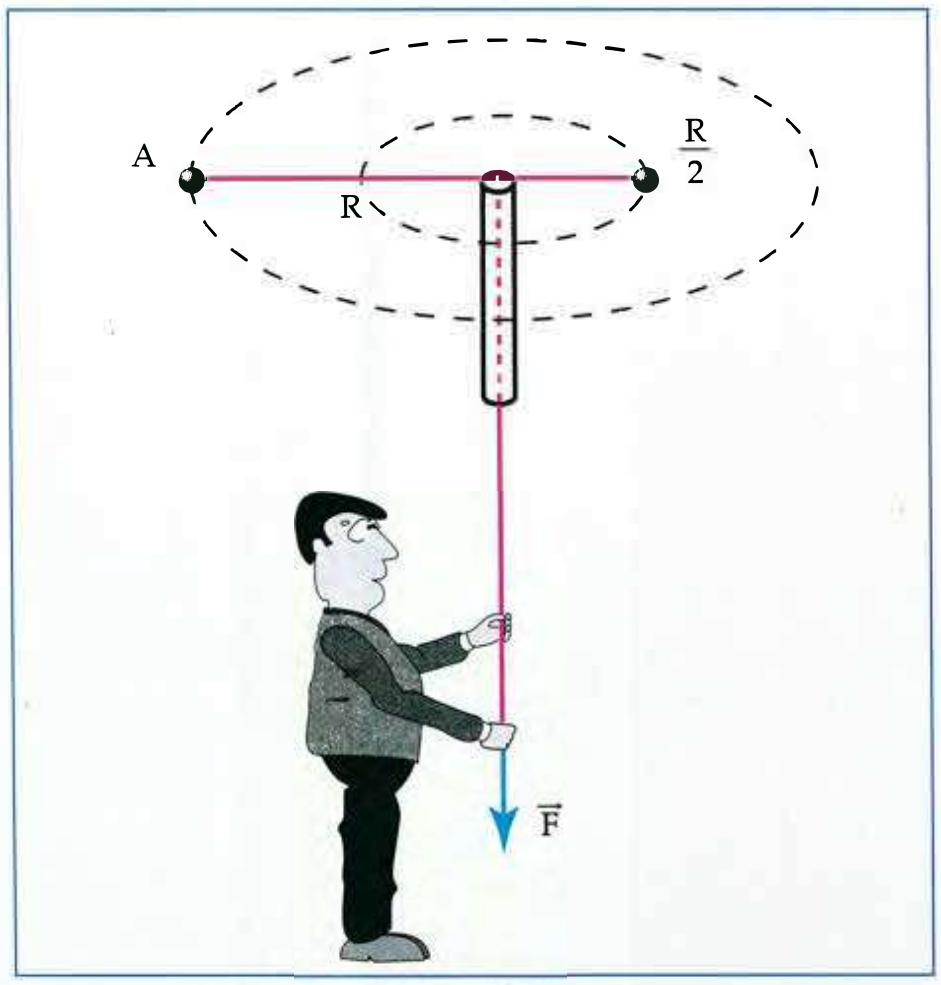

Figure 3.16

a) Give the expression for the magnitude of the force exerted by the observer, in terms of m, R, and θ'_1.
b) The observer pulls the string vertically downward such that the particle describes a smaller circle of radius $r = \frac{R}{2}$. Find the new angular velocity of the particle A in terms of θ'_1.

15 In figure 3.17, the collar C may slide along the vertical axis (Δ) while the two small balls A and B, each of mass m = 500 g, rotate without friction about (Δ) with a constant angular velocity of 4 rd/s. A and B are 15 cm away from (Δ). C is pulled down by a vertical force until the distance between each ball and (Δ) becomes 5 cm. Calculate the work needed to do so bearing in mind that the connecting rods are of negligible masses.

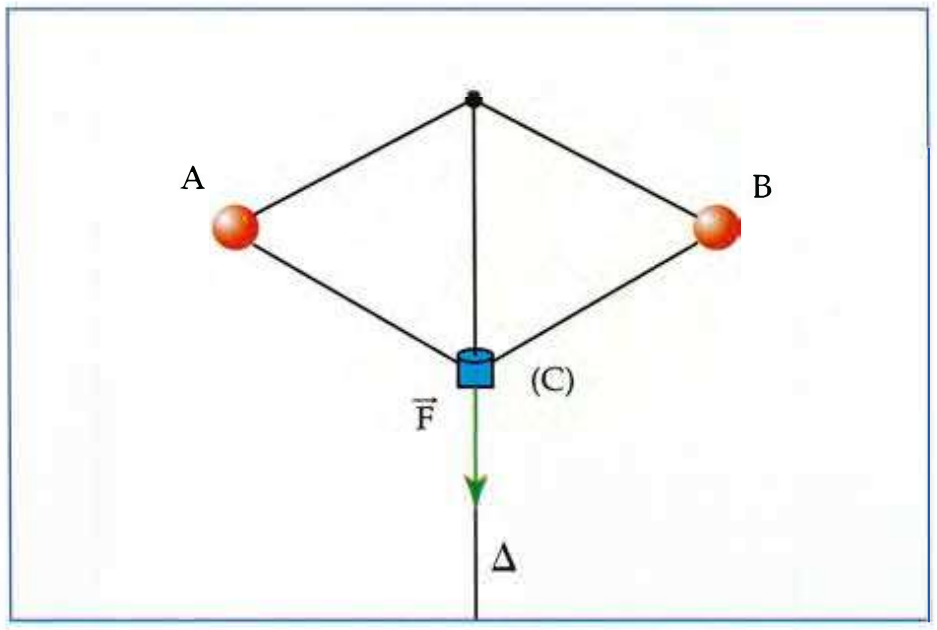

Figure 3.17

16 A man of mass 100 kg stands on the periphery of a horizontal disk of radius 2 m and of moment of inertia with respect to its axis of rotation 4000 kg.m². The system (disk, man) is initially at rest. The man moves along the periphery with a velocity of 1m/s with respect to Earth (Figure 3.18).

a) At what speed and in what direction would the disk rotate?
b) Calculate the angle of rotation followed by the disk when the man takes his initial position with respect to Earth for the first time.

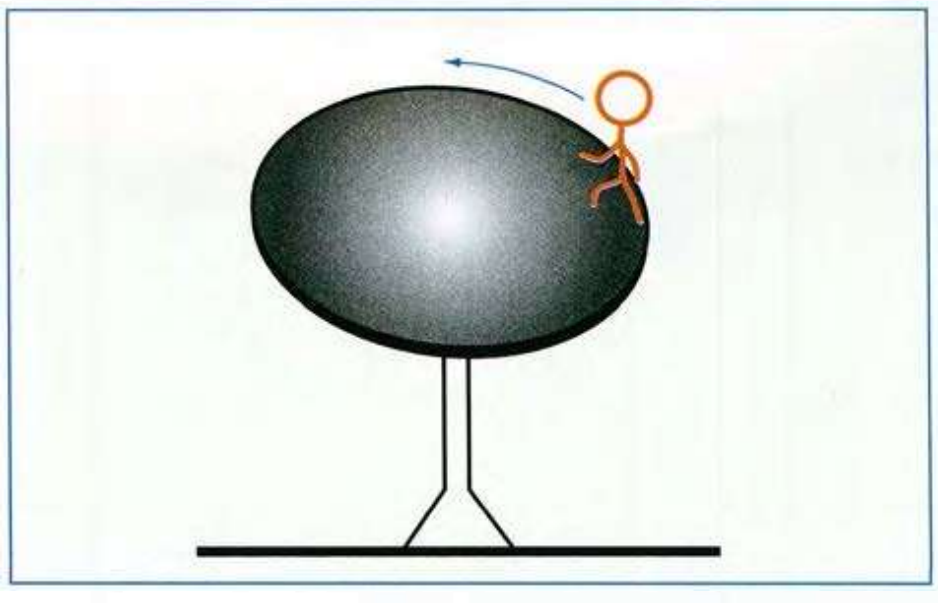

Figure 3.18